

NAME: _____

PERIOD: _____

DATE: _____

A Test of Mean Resolution Time

AP Statistics · Topic 7.5 (CED 4.5) · B: 2027-01-21 · E: 2027-03-08

[STAR] TODAY'S PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE — before the video (5 min)

Which account is more defensible? Cite the p-value comparison or a claim that a test can or cannot justify.

[BOOK] COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)

For a one-sample mean test with unknown population standard deviation, $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ has $df = n - 1$. The p -value is the probability, assuming H_0 is true, of a test statistic at least as extreme as the observed statistic in the direction of H_a . If $p \leq \alpha$, reject H_0 ; if $p > \alpha$, fail to reject H_0 . Failing to reject says the sample does not provide convincing evidence for H_a at that significance level. It does not prove the null value, and the conclusion must concern the population represented by the random sample.

Finish after the video:

DOK 1 (a) Define the population mean and state hypotheses for the research question.

DOK 2 (b) Verify random, 10 percent, and shape conditions. Compute SE , t , df , and the lower-tail p-value; interpret the p-value.

DOK 3 (c) Adjudicate the accounts. Compare p with α , distinguish the formal decision from acceptance or proof, state the scoped population conclusion, and explain the two beliefs produced by the rejected account.

Frame: The numerical comparison is p _____ α , so the formal decision is _____ and the population conclusion is _____.

Frame: That decision _____ establish _____ because the p-value is calculated assuming _____.

EXIT — turn this in

One thing the video changed about my first take: _____